The Wave Kinetic Equation

Yves Peysson, Lorenzo Morini and Joan Decker

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We now derive the kinetic equation for the wave in presence of scattering by density and magnetic field fluctuations. The incident wave is treated as a monochromatic wave, satisfying the homogeneous wave equation. If we have fluctuations of density and magnetic field, $\delta n\left(\widetilde{\mathbf{k}},\widetilde{\omega}\right)$ and $\delta\mathbf{B}\left(\widetilde{\mathbf{k}},\widetilde{\omega}\right)$, from the conservation of momentum and energy, the wave vector and the frequency of the scattered wave are given by the expressions:

$$\check{\mathbf{k}}=\mathbf{k}+\widetilde{\mathbf{k}},\quad \check{\omega}=\omega+\widetilde{\omega} \tag{1}$$

The field of the scattered wave, $\check{\mathbf{E}}\left(\check{\mathbf{k}},\check{\omega}\right)$, satisfies the equation:

$$\nabla\times\nabla\times\check{\mathbf{E}}+\varepsilon_0\mu_0\frac{\partial^2\check{\mathbf{E}}}{\partial \mathrm{t}^2}=-\mu_0\frac{\partial}{\partial \mathrm{t}}\left(\check{\mathbf{J}}+\widetilde{\mathbf{J}}\right) \tag{2}$$

Where $\check{\mathbf{J}}$ is the current due to the linear response of the plasma to the field of the scattered wave, and $\widetilde{\mathbf{J}}$ is the current due to the interaction of the incident wave with the fluctuations. In the Fourier space we have:

$$\check{\mathbf{k}}\times\check{\mathbf{k}}\times\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\check{\omega}\right)+\varepsilon_0\mu_0\check{\omega}^2\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\check{\omega}\right)=-i\mu_0\check{\omega}\left(\check{\mathbf{J}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\check{\omega}\right)+\widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k}.\check{\mathbf{k}},\omega,\check{\omega}\right)\right) \tag{3}$$

So, $\check{\mathbf{J}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\check{\omega}\right)$ is given by:

$$\check{\mathbf{J}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\check{\omega}\right)=\mathbb{S}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\check{\omega}\right) \tag{4}$$

Inserting this expression in the wave equation and introducing the dielectric tensor, we obtain the non homogeneous equation:

$$\check{\mathbf{k}}\times\check{\mathbf{k}}\times\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\check{\omega}\right)+\frac{\check{\omega}^2}{\mathrm{c}^{\,2}}\mathbb{K}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\check{\omega}\right)=-i\mu_0\check{\omega}\widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right) \tag{5}$$

Using the dispersion tensor the equation becomes:

$$\mathbb{D}\left(\check{\mathbf{k}},\check{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\check{\omega}\right)=-\frac{i}{\varepsilon_0\check{\omega}}\widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right) \tag{6}$$

An explicit expression for the current $\widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right)$ is obtained applying the Fourier trasform to the MHD equations, the derivation of this expression

using the linear theory in the cold plasma approximation is discussed in details in appendix. In the cold plasma limit, considering only the component of the current due to the electrons, and several incident modes with wave vector $\mathbf{k}$ and frequency ω, we have:

$$\widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k},\check{\mathbf{k}},\omega,\breve{\omega}\right)=i\varepsilon_0\sum_{\mathbf{k},\omega}\omega\left[\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k},\breve{\omega}-\omega\right)}{n}+\right.$$

$$\left.+i\frac{\omega_{ce}\breve{\omega}}{\omega_{pe}^2}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}},\breve{\omega}\right)\right)\cdot\left(\mathbb{E}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k},\breve{\omega}-\omega\right)}{B_{\mathbf{k}}\left(\mathbf{k},\omega\right)}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\right]\cdot\mathbf{E}_{\mathbf{k}}\left(\mathbf{k},\omega\right) \tag{7}$$

Where $\mathbb{E}^{LC}$ is the Levi-Civita tensor of order 3, and $\mathbb{K}$ is the cold plasma permittivity tensor. Using this expression for the current due to the fluctuations, the wave equation becomes:

$$\mathbb{D}\left(\check{\mathbf{k}},\breve{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}\breve{\omega}\right)=\sum_{\mathbf{k},\omega}\left[\frac{\omega}{\breve{\omega}}\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k},\breve{\omega}-\omega\right)}{n}+\right.$$

$$\left.+i\frac{\omega_{ce}\omega}{\omega_{pe}^2}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}},\breve{\omega}\right)\right)\cdot\left(\mathbb{E}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k},\breve{\omega}-\omega\right)}{B_{\mathbf{k}}\left(\mathbf{k},\omega\right)}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\right]\cdot\mathbf{E}_{\mathbf{k}}\left(\mathbf{k},\omega\right) \tag{8}$$

Fluctuations are assumed to be slow, so we have $\widetilde{\omega}\ll\omega$ and $\breve{\omega}\approx\omega$, as a consequence:

$$\mathbb{D}\left(\check{\mathbf{k}},\breve{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\breve{\omega}\right)=\sum_{\mathbf{k},\omega}\left[\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n}+\right.$$

$$\left.+i\frac{\omega_{ce}\omega}{\omega_{pe}^2}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}},\breve{\omega}\right)\right)\cdot\left(\mathbb{E}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B_{\mathbf{k}}\left(\mathbf{k},\omega\right)}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\right]\cdot\mathbf{E}_{\mathbf{k}}\left(\mathbf{k},\omega\right) \tag{9}$$

Now, we assume that fluctuations of the density and of the magnetic field are sufficienly small to be treated with a perturbative approach, it means that the effects of the fluctuations may be rapresented by a small variation from the normal mode propagation that we can easily study by solving the homogeneous wave equation. In the weak damping approximation, the frequency of the scattered wave is given by the expression:

$$\breve{\omega}=\breve{\omega}_{\check{k}}+i\frac{\partial}{\partial\mathrm{t}}\qquad\text{with}\quad\left|\frac{\partial}{\partial\mathrm{t}}\right|\ll\breve{\omega}_{\check{k}} \tag{10}$$

Where $\breve{\omega}_{\check{k}}$ is the frequency of the normal mode, that is decribed by the homogeneous equation:

$$\mathbb{D}\left(\check{\mathbf{k}},\breve{\omega}_{\check{k}}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\breve{\omega}_{\check{k}}\right)=0 \tag{11}$$

$\breve{\omega}_{\check{k}}$ can be easily determined by solving the dispersion relation associated with the homogeneous equation. Inserting the expression for $\breve{\omega}$ in the wave equation, and expressing the electric field in function of the polarization vector $\mathbf{E}(\mathbf{k})=\mathbf{e}\left(\mathbf{k}\right)E\left(\mathbf{k}\right)$, we obtain:

$$\left[\mathbb{D}\left(\check{\mathbf{k}},\breve{\omega}_{\check{k}}\right)+i\frac{\partial\mathbb{D}}{\partial\breve{\omega}}\frac{\partial}{\partial\mathrm{t}}\right]\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\check{E}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}\right)=\sum_{\mathbf{k}}\left[\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n}+\right.$$

$$\left.+i\frac{\omega_{ce}\omega}{\omega_{pe}^{2}}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}}\right)\right)\cdot\left(\mathbb{E}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B_{\mathbf{k}}\left(\mathbf{k}\right)}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\right]\cdot\mathbf{e}\left(\mathbf{k}\right)E_{\mathbf{k}}\left(\mathbf{k}\right) \tag{12}$$

In this equation all the quantities are considered for $\omega=\breve{\omega}_{\check{k}}$. Multiplying this equation for $\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)$ and remembering that $\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{D}\left(\check{\mathbf{k}},\breve{\omega}_{\check{k}}\right)=0$, the equation finally becomes:

$$i\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\frac{\partial\mathbb{D}}{\partial\breve{\omega}}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\frac{\partial\check{E}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}\right)}{\partial\mathrm{t}}=\sum_{\mathbf{k}}\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\left[\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n}+\right.$$

$$\left.+i\frac{\omega_{ce}\omega}{\omega_{pe}^{2}}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}}\right)\right)\cdot\left(\mathbb{E}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B_{\mathbf{k}}\left(\mathbf{k}\right)}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\right]\cdot\mathbf{e}\left(\mathbf{k}\right)E_{\mathbf{k}}\left(\mathbf{k}\right) \tag{13}$$

In order to study the coupling between the incident and the scattered wave, we introduce the wave energy density:

$$W\left(\mathbf{k}\right)=\varepsilon_{0}\frac{\breve{\omega}_{\check{k}}}{2}\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\frac{\partial\mathbb{D}}{\partial\omega}\cdot\mathbf{e}\left(\mathbf{k}\right)\left|\mathbf{E}\left(\mathbf{k}\right)\right|^{2}=\varepsilon_{0}\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)E\left(\mathbf{k}\right)^{2} \tag{14}$$

Where $\mathbb{M}=\frac{\breve{\omega}_{\check{k}}}{2}\frac{\partial\mathbb{D}}{\partial\omega}$. The wave amplitude is given by:

$$C\left(\mathbf{k}\right)=W\left(\mathbf{k}\right)^{\frac{1}{2}} \tag{15}$$

So we exprime the electric field in function of the wave amplitude, and we obtain the mode coupling equation:

$$i\frac{\partial C\left(\check{\mathbf{k}}\right)}{\partial\mathrm{t}}=\sum_{\mathbf{k}}\frac{\breve{\omega}_{\check{k}}}{2}\left[\frac{\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\cdot\mathbf{e}\left(\mathbf{k}\right)}{\left(\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}}\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n}+\right.$$

$$+i\frac{\omega_{ce}\check{\omega}_{\check{k}}}{\omega_{pe}^2}\frac{\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}}\right)\right)\cdot\left(\mathbb{E}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B_{\mathbf{k}}(\mathbf{k})}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\cdot\mathbf{e}\left(\mathbf{k}\right)}{\left(\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}}\Bigg]C\left(\mathbf{k}\right) \tag{16}$$

This equation can be written in the form:

$$i\frac{\partial C\left(\check{\mathbf{k}}\right)}{\partial \mathrm{t}}=\sum_{\mathbf{k}}\left[V^{n}\left(\mathbf{k},\check{\mathbf{k}},\check{\omega}_{\check{k}}\right)\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n}+\mathbf{V}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\check{\omega}_{\check{k}}\right)\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B_{\mathbf{k}}}\right]C\left(\mathbf{k}\right) \tag{17}$$

Where $V^{n}\left(\mathbf{k},\check{\mathbf{k}},\omega_{k}\right)$and $\mathbf{V}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega_{k}\right)$ are the coupling coefficients:

$$V^{n}\left(\mathbf{k},\check{\mathbf{k}},\omega_{k}\right)=\frac{\check{\omega}_{\check{k}}}{2}\frac{\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\cdot\mathbf{e}\left(\mathbf{k}\right)}{\left(\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \tag{18}$$

$$\mathbf{V}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega_{k}\right)=\frac{i\omega_{ce}\check{\omega}_{\check{k}}^{2}}{2\omega_{pe}^{2}}\frac{\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}}\right)\right)\cdot\mathbb{E}^{LC}\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\cdot\mathbf{e}\left(\mathbf{k}\right)}{\left(\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \tag{19}$$

Using the component notation:

$$V^{n}\left(\mathbf{k},\check{\mathbf{k}},\omega_{k}\right)=\frac{\check{\omega}_{\check{k}}}{2}\frac{\check{e}_{i}^{\star}\left(\check{\mathbf{k}}\right)\left(\delta_{ij}-K_{ij}\left(\mathbf{k}\right)\right)e_{j}\left(\mathbf{k}\right)}{\left(\check{e}_{i}^{\star}\left(\check{\mathbf{k}}\right)M_{ij}\check{e}_{j}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(e_{k}^{\star}\left(\mathbf{k}\right)M_{kl}e_{l}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \tag{20}$$

$$V_{m}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega_{k}\right)=\frac{i\omega_{ce}\check{\omega}_{\check{k}}^{2}}{2\omega_{pe}^{2}}\frac{\check{e}_{i}^{\star}\left(\check{\mathbf{k}}\right)\left(\delta_{il}-K_{il}\left(\check{\mathbf{k}}\right)\right)\epsilon_{lkm}\left(\delta_{kj}-K_{kj}\left(\mathbf{k}\right)\right)e_{j}\left(\mathbf{k}\right)}{\left(\check{e}_{i}^{\star}\left(\check{\mathbf{k}}\right)M_{ij}\check{e}_{j}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(e_{k}^{\star}\left(\mathbf{k}\right)M_{kl}e_{l}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \tag{21}$$

Appendix A

We now derive the explicit expression for the current $\widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right)$ due to interaction between the field of the incident wave and the fluctuation of the density and of the magnetic field, in the cold plasma approximation.

We firstly assume to have only fluctuations of the density, with $\widetilde{\omega}\ll\omega$, so $\check{\omega}=\omega$. In the cold plasma model, the effects of the thermal motion of the particles and of the collisions are neglected, each species of particle is treated as an indipendent fluid, so for studying wave propagation we need to solve the Maxwell equations coupled with the equation of the motion for the various species of particles:

$$\nabla\times\mathbf{E}=-\frac{\partial\mathbf{B}}{\partial \mathrm{t}} \tag{22}$$

$$\nabla\times\mathbf{B}=\varepsilon_{0}\mu_{0}\frac{\partial\mathbf{E}}{\partial \mathrm{t}}+\mu_{0}\mathbf{J} \tag{23}$$

$$\mathrm{m}_s\left(\frac{\partial \mathbf{v}_s}{\partial \mathrm{t}}+\left(\mathbf{v}_s\nabla\right)\mathbf{v}_s\right)=\mathrm{q}_s\left(\mathbf{E}+\frac{\mathbf{v}_s}{\mathrm{c}}\times\mathbf{B}\right) \tag{24}$$

Where the current $\mathbf{J}$ is given by the expression:

$$\mathbf{J}=\sum_s \mathrm{q}_s n_s \mathbf{v}_s \tag{25}$$

Since the difference between electron and ion masses is very large, and $\mathbf{v}_i \ll \mathbf{v}_e$, we consider only the component of the current due to the electrons, assuming $n_e = n$ and introducing the density fluctuations, we have:

$$\mathbf{J}=-e\left(n+\delta n\right)\mathbf{v}_e \tag{26}$$

Neglecting the convective term, the linearized equation of motion, in absence of magnetic field and velocity fluctuation, is:

$$\mathrm{m}_e\frac{\partial \mathbf{v}_e}{\partial \mathrm{t}}=-e\left(\mathbf{E}+\frac{\mathbf{v}_e}{\mathrm{c}}\times\mathbf{B}\right) \tag{27}$$

In the Fourier space we have:

$$i\mathrm{m}_e\omega\left[\mathbf{v}_{e\mathbf{k}}\left(\mathbf{k},\omega\right)+i\frac{\omega_{ce}}{\omega}\mathbf{v}_{e\mathbf{k}}\left(\mathbf{k},\omega\right)\times\mathbf{b}_{\mathbf{k}}\left(\mathbf{k},\omega\right)\right]=e\mathbf{E}\left(\mathbf{k},\omega\right) \tag{28}$$

Using the tensorial notation:

$$i\mathrm{m}_e\omega\left(\mathbb{I}+i\frac{\omega_{ce}}{\omega}\mathbb{E}^{LC}\cdot\mathbf{b}_{\mathbf{k}}\left(\mathbf{k},\omega\right)\right)\cdot\mathbf{v}_{e\mathbf{k}}\left(\mathbf{k},\omega\right)=e\mathbf{E}\left(\mathbf{k},\omega\right) \tag{29}$$

So the electron velocity is given by the expression:

$$\mathbf{v}_{e\mathbf{k}}\left(\mathbf{k},\omega\right)=-\frac{ie}{\mathrm{m}_e\omega}\mathbb{T}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}\left(\mathbf{k},\omega\right) \tag{30}$$

Where the tensor $\mathbb{T}$ is:

$$\mathbb{T}\left(\mathbf{k},\omega\right)=\frac{1}{\omega^2-\omega_{ce}^2}\left(\omega^2\mathbb{I}-\omega_{ce}^2\mathbf{b}_{\mathbf{k}}\mathbf{b}_{\mathbf{k}}-i\omega\omega_{ce}\mathbb{E}^{LC}\cdot\mathbf{b}_{\mathbf{k}}\right)$$

Using (30) in (25), the current becomes:

$$\mathbf{J}_{\mathbf{k}}=\frac{ie^2n}{\mathrm{m}_e\omega}\left(1+\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n}\right)\mathbb{T}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}\left(\mathbf{k},\omega\right)=\frac{i\omega_{pe}^2\varepsilon_0}{\omega}\left(1+\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n}\right)\mathbb{T}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}\left(\mathbf{k},\omega\right) \tag{31}$$

So the component due to the interaction between the field of the incident wave and the fluctuations is:

$$\widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right)=\frac{i\omega_{pe}^2\varepsilon_0}{\omega}\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n}\mathbb{T}\left(\mathbf{k},\omega\right)\cdot\mathbf{E}_{\mathbf{k}}\left(\mathbf{k},\omega\right) \tag{32}$$

In absence of fluctuations, we have:

$$\mathbf{J}_{\mathbf{k}}(\mathbf{k},\omega) = \mathbb{S}(\mathbf{k},\omega)\cdot\mathbf{E}_{\mathbf{k}}(\mathbf{k},\omega) = i\omega\varepsilon_0(\mathbb{I}-\mathbb{K}(\mathbf{k},\omega))\cdot\mathbf{E}_{\mathbf{k}}(\mathbf{k},\omega) = \frac{i\omega_{pe}^2\varepsilon_0}{\omega}\mathbb{T}(\mathbf{k},\omega)\cdot\mathbf{E}_{\mathbf{k}}(\mathbf{k},\omega) \tag{33}$$

Where $\mathbb{S}$ and $\mathbb{K}$ are respectively the conductivity and the permittivity tensor of a cold plasma. From last equation we can deduce an expression of $\mathbb{T}$ in function of the cold plasma permittivity tensor:

$$\mathbb{T}(\mathbf{k},\omega) = \frac{\omega^2}{\omega_{pe}^2}(\mathbb{I}-\mathbb{K}(\mathbf{k},\omega))$$

So the current due to the fluctuation of the density can be finally expressed in the form:

$$\widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right) = i\omega\varepsilon_0\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n}(\mathbb{I}-\mathbb{K}(\mathbf{k},\omega))\cdot\mathbf{E}_{\mathbf{k}}(\mathbf{k},\omega) \tag{34}$$

In components:

$$\widetilde{J}_{\mathbf{k},\check{\mathbf{k}},i} = i\omega\varepsilon_0(\delta_{ij}-K_{ij}(\mathbf{k},\omega))\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n}E_{\mathbf{k},j} \tag{35}$$

Appendix B

We now calculate explicitely the mode coupling coefficient for the density fluctuations $V^n\left(\mathbf{k},\check{\mathbf{k}},\omega\right)$:

$$V^n\left(\mathbf{k},\check{\mathbf{k}},\omega\right) = \frac{\omega}{2}\frac{\check{\mathbf{e}}^\star\left(\check{\mathbf{k}}\right)\cdot(\mathbb{I}-\mathbb{K}(\mathbf{k}))\cdot\mathbf{e}(\mathbf{k})}{\left(\check{\mathbf{e}}^\star\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\cdot\left(\mathbf{e}^\star(\mathbf{k})\cdot\mathbb{M}\cdot\mathbf{e}(\mathbf{k})\right)^{\frac{1}{2}}} \tag{36}$$

The numerator of the fraction is given by the product:

$$\check{\mathbf{e}}^\star\left(\check{\mathbf{k}}\right)\cdot(\mathbb{I}-\mathbb{K}(\mathbf{k}))\cdot\mathbf{e}(\mathbf{k}) =$$

$$= \check{e}_x^\star e_x(1-K_{xx}) + \check{e}_y^\star e_x K_{yx} + \check{e}_z^\star e_x K_{zx} + \check{e}_x^\star e_y K_{xy} + \check{e}_y^\star e_y(1-K_{yy}) + \check{e}_z^\star e_y K_{zy} +$$

$$+\check{e}_x^\star e_z K_{xz} + \check{e}_y^\star e_z K_{yz} + \check{e}_z^\star e_z(1-K_{zz}) =$$

$$= \check{e}_x^\star\left[e_x(1-K_{xx}) + e_y K_{xy} + e_z K_{xz}\right] + \check{e}_y^\star\left[e_x K_{yx} + e_y(1-K_{yy}) + e_z K_{yz}\right] +$$

$$+\check{e}_z^\star\left[e_x K_{zx} + e_y K_{zy} + e_z(1-K_{zz})\right] \tag{37}$$

The two factors of the denominator are:

$$\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)=$$

$$=\check{e}_x^{\star}\check{e}_xM_{xx}+\check{e}_y^{\star}\check{e}_xM_{yx}+\check{e}_z^{\star}\check{e}_xM_{zx}+\check{e}_x^{\star}\check{e}_yM_{xy}+\check{e}_y^{\star}\check{e}_yM_{yy}+\check{e}_z^{\star}\check{e}_yM_{zx}+$$

$$+\check{e}_x^{\star}\check{e}_zM_{xz}+\check{e}_y^{\star}\check{e}_zM_{yz}+\check{e}_z^{\star}\check{e}_zM_{zz}=$$

$$=\check{e}_x^{\star}\left[\check{e}_xM_{xx}+\check{e}_yM_{xy}+\check{e}_zM_{xz}\right]+\check{e}_y^{\star}\left[\check{e}_xM_{yx}+\check{e}_yM_{yy}+\check{e}_zM_{yz}\right]+$$

$$+\check{e}_z^{\star}\left[\check{e}_xM_{zx}+\check{e}_yM_{zy}+\check{e}_zM_{zz}\right] \tag{38}$$

$$\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)=$$

$$=e_x^{\star}e_xM_{xx}+e_y^{\star}e_xM_{yx}+e_z^{\star}e_xM_{zx}+e_x^{\star}e_yM_{xy}+e_y^{\star}e_yM_{yy}+e_z^{\star}e_yM_{zy}+$$

$$+e_x^{\star}e_zM_{xz}+e_y^{\star}e_zM_{yz}+e_z^{\star}e_zM_{zz}=$$

$$=e_x^{\star}\left[e_xM_{xx}+e_yM_{xy}+e_zM_{xz}\right]+e_y^{\star}\left[e_xM_{yx}+e_yM_{yy}+e_zM_{yz}\right]+$$

$$+e_z^{\star}\left[e_xM_{zx}+e_yM_{zy}+e_zM_{zz}\right] \tag{39}$$

We need an expression for the tensor $\mathbb{M}$ and for the polarization vector. Remembering that the dispersion tensor is given by:

$$\mathbb{D}=\begin{pmatrix} K_{xx}-N_{\parallel}^2 & K_{xy} & K_{xz}+N_{\perp}N_{\parallel} \\ -K_{xy} & K_{yy}-N^2 & K_{yz} \\ K_{xz}+N_{\perp}N_{\parallel} & -K_{yz} & K_{zz}-N_{\perp}^2 \end{pmatrix} \tag{40}$$

The elements of the tensor $\mathbb{M}$ are:

$$
\begin{aligned}
M_{xx} = \quad \tfrac{\omega}{2}\tfrac{\partial D_{xx}}{\partial \omega} &= \quad \frac{\omega}{2}\frac{\partial K_{xx}}{\partial \omega} + N_{\parallel}^2 \\
M_{xy} = \quad \tfrac{\omega}{2}\tfrac{\partial D_{xy}}{\partial \omega} &= \quad \frac{\omega}{2}\frac{\partial K_{xy}}{\partial \omega} \\
M_{xz} = \quad \tfrac{\omega}{2}\tfrac{\partial D_{xz}}{\partial \omega} &= \quad \frac{\omega}{2}\frac{\partial K_{xz}}{\partial \omega} - N_{\perp}N_{\parallel} \\
M_{yx} = \quad \tfrac{\omega}{2}\tfrac{\partial D_{yx}}{\partial \omega} &= \quad -\frac{\omega}{2}\frac{\partial K_{xy}}{\partial \omega} \\
M_{yy} = \quad \tfrac{\omega}{2}\tfrac{\partial D_{yy}}{\partial \omega} &= \quad \frac{\omega}{2}\frac{\partial K_{yy}}{\partial \omega} + N^2 \\
M_{yz} = \quad \tfrac{\omega}{2}\tfrac{\partial D_{yz}}{\partial \omega} &= \quad \frac{\omega}{2}\frac{\partial K_{yz}}{\partial \omega} \\
M_{zx} = \quad \tfrac{\omega}{2}\tfrac{\partial D_{zx}}{\partial \omega} &= \quad \frac{\omega}{2}\frac{\partial K_{zx}}{\partial \omega} - N_{\perp}N_{\parallel} \\
M_{zy} = \quad \tfrac{\omega}{2}\tfrac{\partial D_{zy}}{\partial \omega} &= \quad -\frac{\omega}{2}\frac{\partial K_{yz}}{\partial \omega} \\
M_{zz} = \quad \tfrac{\omega}{2}\tfrac{\partial D_{zz}}{\partial \omega} &= \quad \frac{\omega}{2}\frac{\partial K_{zz}}{\partial \omega} + N_{\perp}^2
\end{aligned}
$$

So the tensor $\mathbb{M}$ is:

$$
\mathbb{M} = \frac{\omega}{2}\frac{\partial \mathbb{D}}{\partial \omega} = \begin{pmatrix} \frac{\omega}{2}\frac{\partial K_{xx}}{\partial \omega} + N_{\parallel}^2 & \frac{\omega}{2}\frac{\partial K_{xy}}{\partial \omega} & \frac{\omega}{2}\frac{\partial K_{xz}}{\partial \omega} - N_{\perp}N_{\parallel} \\ -\frac{\omega}{2}\frac{\partial K_{xy}}{\partial \omega} & \frac{\omega}{2}\frac{\partial K_{yy}}{\partial \omega} + N^2 & \frac{\omega}{2}\frac{\partial K_{yz}}{\partial \omega} \\ \frac{\omega}{2}\frac{\partial K_{xz}}{\partial \omega} - N_{\perp}N_{\parallel} & -\frac{\omega}{2}\frac{\partial K_{yz}}{\partial \omega} & \frac{\omega}{2}\frac{\partial K_{zz}}{\partial \omega} + N_{\perp}^2 \end{pmatrix} \tag{41}
$$

The derivatives of the elements of the permittivity tensor $\mathbb{K}\left(\mathbf{N}, \beta_{Ts}, \overline{\omega}_{ps}, \overline{\omega}_{cs}\right)$ respect to the frequency ω are given by the expressions:

$$
\begin{aligned}
\frac{\partial K_{xx}}{\partial \omega} &= \frac{\partial K_{xx}}{\partial \overline{\omega}_{ps}}\frac{\partial \overline{\omega}_{ps}}{\partial \omega} + \frac{\partial K_{xx}}{\partial \overline{\omega}_{cs}}\frac{\partial \overline{\omega}_{cs}}{\partial \omega} = -\frac{\partial K_{xx}}{\partial \overline{\omega}_{ps}}\frac{\overline{\omega}_{ps}}{\omega} - \frac{\partial K_{xx}}{\partial \overline{\omega}_{cs}}\frac{\overline{\omega}_{cs}}{\omega} \\
\frac{\partial K_{xy}}{\partial \omega} &= \frac{\partial K_{xy}}{\partial \overline{\omega}_{ps}}\frac{\partial \overline{\omega}_{ps}}{\partial \omega} + \frac{\partial K_{xy}}{\partial \overline{\omega}_{cs}}\frac{\partial \overline{\omega}_{cs}}{\partial \omega} = -\frac{\partial K_{xy}}{\partial \overline{\omega}_{ps}}\frac{\overline{\omega}_{ps}}{\omega} - \frac{\partial K_{xy}}{\partial \overline{\omega}_{cs}}\frac{\overline{\omega}_{cs}}{\omega} \\
\frac{\partial K_{xz}}{\partial \omega} &= \frac{\partial K_{xz}}{\partial \overline{\omega}_{ps}}\frac{\partial \overline{\omega}_{ps}}{\partial \omega} + \frac{\partial K_{xz}}{\partial \overline{\omega}_{cs}}\frac{\partial \overline{\omega}_{cs}}{\partial \omega} = -\frac{\partial K_{xz}}{\partial \overline{\omega}_{ps}}\frac{\overline{\omega}_{ps}}{\omega} - \frac{\partial K_{xz}}{\partial \overline{\omega}_{cs}}\frac{\overline{\omega}_{cs}}{\omega} \\
\frac{\partial K_{yy}}{\partial \omega} &= \frac{\partial K_{yy}}{\partial \overline{\omega}_{ps}}\frac{\partial \overline{\omega}_{ps}}{\partial \omega} + \frac{\partial K_{yy}}{\partial \overline{\omega}_{cs}}\frac{\partial \overline{\omega}_{cs}}{\partial \omega} = -\frac{\partial K_{yy}}{\partial \overline{\omega}_{ps}}\frac{\overline{\omega}_{ps}}{\omega} - \frac{\partial K_{yy}}{\partial \overline{\omega}_{cs}}\frac{\overline{\omega}_{cs}}{\omega} \\
\frac{\partial K_{yz}}{\partial \omega} &= \frac{\partial K_{yz}}{\partial \overline{\omega}_{ps}}\frac{\partial \overline{\omega}_{ps}}{\partial \omega} + \frac{\partial K_{yz}}{\partial \overline{\omega}_{cs}}\frac{\partial \overline{\omega}_{cs}}{\partial \omega} = -\frac{\partial K_{yz}}{\partial \overline{\omega}_{ps}}\frac{\overline{\omega}_{ps}}{\omega} - \frac{\partial K_{yz}}{\partial \overline{\omega}_{cs}}\frac{\overline{\omega}_{cs}}{\omega} \\
\frac{\partial K_{zz}}{\partial \omega} &= \frac{\partial K_{zz}}{\partial \overline{\omega}_{ps}}\frac{\partial \overline{\omega}_{ps}}{\partial \omega} + \frac{\partial K_{zz}}{\partial \overline{\omega}_{cs}}\frac{\partial \overline{\omega}_{cs}}{\partial \omega} = -\frac{\partial K_{zz}}{\partial \overline{\omega}_{ps}}\frac{\overline{\omega}_{ps}}{\omega} - \frac{\partial K_{zz}}{\partial \overline{\omega}_{cs}}\frac{\overline{\omega}_{cs}}{\omega}
\end{aligned}
$$

The kernel of the scattering probability is given by the expression:

$$K\left(k_{\perp},\check{k}_{\perp},\beta\right)=\frac{2\pi\check{k}_{\perp}}{v_{g\perp}}\left|V^{n}\left(\mathbf{k},\check{\mathbf{k}},\beta\right)\right|^{2} \tag{42}$$

The group velocity is given by the expression:

$$\mathbf{v}_{g}=\frac{\partial\omega}{\partial\mathbf{k}}=-\frac{\partial\mathcal{D}/\partial\mathbf{k}}{\partial\mathcal{D}/\partial\omega}=-\frac{\mathrm{c}}{\omega}\frac{\partial\mathcal{D}/\partial\mathbf{N}}{\partial\mathcal{D}/\partial\omega} \tag{43}$$

Where $\mathcal{D}$ is the dispersio relation, the component perpendicular to the magnetic field is:

$$v_{g\perp}=-\frac{\partial\mathcal{D}/\partial k_{\perp}}{\partial\mathcal{D}/\partial\omega}=-\frac{\mathrm{c}}{\omega}\frac{\partial\mathcal{D}/\partial N_{\perp}}{\partial\mathcal{D}/\partial\omega} \tag{44}$$

So the probability kernel becomes:

$$K\left(k_{\perp},\check{k}_{\perp},\beta\right)=-2\pi k_{\perp}\frac{\partial\mathcal{D}/\partial\omega}{\partial\mathcal{D}/\partial k_{\perp}}\left|V^{n}\left(\mathbf{k},\check{\mathbf{k}},\beta\right)\right|^{2}=-\frac{2\pi\omega^{2}}{\mathrm{c}^{\,2}}N_{\perp}\frac{\partial\mathcal{D}/\partial\omega}{\partial\mathcal{D}/\partial N_{\perp}}\left|V^{n}\left(\mathbf{k},\check{\mathbf{k}},\beta\right)\right|^{2} \tag{45}$$